

June (Junhyeok) Lee

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SUMMARY

I develop clinically aligned representation learning systems for biomedical AI. I move beyond black-box optimization by embedding physiological and structural priors into models to improve robustness under real-world distribution shift.

EDUCATION

2025 – Present	M.S. in Bioengineering , University of Washington Focus: Multi-scale biomechanics and physics-informed deep learning <i>Expected Graduation: June 2026</i>	GPA: 3.9/4.0
2018 – 2024	B.S. in Biomedical Engineering , Hankuk University of Foreign Studies Thesis: <i>Deep Clustering of Single-cell RNA-seq with Combined Features</i>	GPA: 3.9/4.0

WORK EXPERIENCE

Research Scientist — Monitor Corporation	Jul 2023 – Jun 2025
– Probe-Conditional Ultrasound Denoising (Production DDPM): - Deployed contrastive learning with a probe-conditional DDPM to suppress spectral artifacts across three GE ultrasound probe variants in production - Explicitly conditioned the generative model on probe-specific prior distributions to address hardware-induced domain shift, achieving 89% false positive reduction across 50K+ clinical scans - Eliminated per-probe retraining by unifying domain adaptation into a single conditional generative architecture	
– Semi-Supervised 3D Segmentation Pipeline: - Built a Fast Fourier Convolution (FFC) 3D U-Net with multi-scale frequency filtering, achieving 0.72 Dice and sub-second inference for malignant/benign tumor segmentation - Designed a MedSAM-inspired semi-supervised framework with reliability scoring, scaling 300 manually annotated scans to 21,900 auto-labeled scans (1:73 ratio) - Reduced annotation cost by 21,600 hours (\$540K at \$25/hr) while maintaining parity with fully supervised baselines	
– Model Compression for Edge Deployment: - Applied bfloat16 quantization and structured pruning to compress 3D segmentation models by 50% without degrading structural prediction fidelity - Achieved 2.5× faster inference on hospital workstations while preserving 98% accuracy relative to the FP32 baseline, enabling real-time clinical deployment	

RESEARCH EXPERIENCE

ZERO: Zero-shot Effect-vector from Frozen Segmentation Encoders — Asan Hospital	Jan 2025 – Present
- Discovered that frozen segmentation encoders harbor latent biomarker axes recoverable with zero trainable parameters via centroid-difference projections - Achieved directional stability (5-fold $\cos\theta=0.90/0.96$) and zero-shot generalization to held-out patients (CVS AUC 0.803, PRL AUC 0.716) - Validated physical grounding with imaging gradients (SWI gradient $p < 0.001$; PRL rim $\rho = -0.49, p < 0.001$) - Submitted to <i>MICCAI 2026</i> (2nd author)	
WiseFind: Structure-as-Context Multimodal RAG — Harborview Medical Center (HMC)	Sep 2025 – Present
- Identified intra-domain visual homogeneity as a structural domain property (silhouette < 0.06 post-fine-tuning), motivating text- and structure-based retrieval over visual embeddings - Proposed Shotgun Retrieval via THC Tree with three complementary channels (T2I, CI, XRef), achieving P@1 0.848, MRR 0.872, R@5 0.701 on 200 expert neurosurgical queries—3.3× above best visual baseline - Fine-tuned ColBERT (PubMedBERT backbone, 30K steps) on 5,544 domain-specific pairs; XRef channel raises hard-query P@1 by +76% relative for figures > 10 pages from text anchor - Deployed at HMC integrating MedGemma + RAG for real-world neurosurgical workflow testing; submitted to <i>MICCAI 2026</i> (1st author)	

WiseSpine: Physics-Guided Counterfactual Surgical Simulation — HMC

Sep 2025 – Present

- Built a grid-based stress transfer simulator (P2G/G2P, 64^3 grid, 50-step Jacobi diffusion) enabling physically consistent load propagation and fracture isolation
- Implemented anisotropic Continuum Damage Mechanics with $(1 - D)^2$ stiffness degradation and COD modeling, reproducing AO-type fracture evolution (A1–A4)
- Mapped 50K particle-level damage to CT HU space with mass/volume conservation checks
- Formulated segmentation robustness as a physics-constrained PPO min–max game with validated causal DAG analysis (KS-tests, SSIM coherence); preparing *NeurIPS 2026*

Culture-Positive TB Detection and Physiologically Grounded AI Modeling — Ubicomp Lab

Sep 2025 – Present

- Developed an acoustic TB detection system achieving **AUC 0.9519** and **92.7% sensitivity**, reducing diagnostic latency from weeks to real-time screening
- Extracted **57 physiological biomarkers** from 7,031 cough segments, identifying infection-linked spectral and temporal shifts (e.g., 500–3000 Hz resonance)
- Revealed demographic and bacterial-load heterogeneity (female-specific slope $d = 0.593$; age-dependent reversals)
- Demonstrated interpretability via saliency concentration (25.9% in 500–3000 Hz) and biomarker correlation ($|r|$ up to 0.42); In preparation for *Nature Communications*

Tissue-Aware Drug Response Prediction — Bioinformatics Lab

Jan 2022 – Dec 2025

- Modeled tissue identity as a structural prior in pharmacogenomic learning using 54,837 GTEx samples
- Achieved $R^2 = 0.857 \pm 0.012$ on 257,955 drug–cell line pairs with **61.6% MSE reduction**
- Designed cross-attention multimodal architecture with biologically validated gene-importance estimation (DrugBank/KEGG $p < 10^{-8}$; $r = 0.40$)
- Manuscript under review at *Bioinformatics*

MRI Bias Field Correction with Generative AI — MedicalVision Lab

Aug 2022 – Apr 2023

- Developed diffusion-based MRI bias field correction with Hampel mixture noise modeling
- Achieved **9.75× faster inference** than standard DDPM while preserving diagnostic fidelity
- Published at **MIDL 2023**

HACKATHON

Google MedGemma 2026 Healthcare Hackathon ([GitHub](#), [Demo](#))

Jan 2026 – Present

- Architected a **three-agent clinical reasoning pipeline** — Audio (**HeAR**), CXR (**MedSigLIP**), and CT (**MedSigLIP**) agents independently analyze complementary modalities, then fuse evidence through confidence-weighted late fusion, mirroring the multi-specialist case conference model
- Designed a **3-level hierarchical reasoning framework** (pathophysiology → pattern recognition → disease-specific biomarkers) with explicit quantitative thresholds — producing a *biological story* explaining *why* findings map to diagnoses, not just *that* they do (e.g., audio silence ↔ CXR hyperlucency ↔ pleural air trapping)
- Formalized **disagreement as a clinical signal**: when agents diverge (e.g., audio predicts Pneumonia while CXR/CT predict COVID-19), the system raises a *NEEDS CONFIRMATION* flag, applies modality-specific weights (CT $\times 1.2$, CXR $\times 1.0$, Audio $\times 0.8$), and surfaces co-morbidity hypotheses rather than silently overriding minority agents
- Designed for **modular deployment in resource-limited settings** — audio-only screening on a smartphone (HeAR), gracefully scaling to CXR then CT; full 3-modality pipeline runs end-to-end in $\sim 5s$ on consumer hardware with FHIR-compatible JSON outputs; submitted to *MedGemma Impact Challenge 2026* (Kaggle)

Seoul National University Hospital (SNUH) Skin Burn Classification Hackathon — 3rd Place

2022

- Developed a two-stage detection + severity classification pipeline for burn depth assessment
- Introduced an attenuation-inspired RGB reweighting scheme (0.21, 0.72, 0.07) to emphasize hemoglobin-sensitive green-channel information prior to model input
- Improved classification performance without increasing model complexity, demonstrating that physiologically motivated preprocessing can outperform heavier architectural modifications
- Awarded **3rd Place** among national-level medical AI teams

PUBLICATIONS

Lee, Junhyeok, Alex Ching*, Cynthia Dong*, David Horne, Thomas Hawn, Shwetak Patel. (2026). Rapid Identification of Culture-Positive Tuberculosis from Cough Sounds Using Deep Learning. *Nature Communications* (Manuscript in preparation).

Lee, Junhyeok, Aruna Souri*, Mary Nguyen*, Benjamin Han*, Christopher W. Lewis, Diana Wiseman. (2026). WiseFind: Why Structure Matters for Traceable Retrieval in Neurosurgery. *Medical Image Computing and Computer Assisted Intervention (MICCAI)* (under review).

Jeong, Eunseon, **Lee, Junhyeok**, Yangshin Choi, Yoonho Nam. (2026). ZERO : Zero-shot Effect-vector Recovery from frOzen segmentation encoders. *Medical Image Computing and Computer Assisted Intervention (MICCAI)* (under review).

Lee, Junhyeok, Jiwon Seo, Younhee Ko. (2026). Interpretable Tissue-Aware Drug Response Prediction Through Biologically Grounded Multimodal Deep Learning. *Bioinformatics* (under review).

Lee, Junhyeok*, J. Kang*, Y. Nam, and T. Lee. (Apr. 2023). Bias Field Correction in MRI with Hampel Noise Denoising Diffusion Probabilistic Model. *Proceedings of Medical Imaging with Deep Learning (MIDL)*.

TEACHING EXPERIENCE

Teaching Assistant — BIOE 437/537 Computational Systems Biology

Fall 2025

University of Washington, Prof. Herbert M. Sauro & Prof. Joseph L. Hellerstein

- Assisting enzyme kinetic simulation and optimization coursework; leading office hours and grading
- Contributing to novel research integrating simulation-generated data for AI training

TECHNICAL SKILLS

Programming Languages	Python, R, Matlab, SQL, C/C++, Flask, Neo4j, MongoDB, ChromaDB
ML/AI Frameworks	PyTorch, Transformers, DDPM, GenAI, AWS SageMaker, LLM, Reasoning, CoT
Systems/Infrastructure	PyCuda, TensorRT, Distributed Training, Docker, SLURM, HPC Cluster, Apptainer
Project Management	Atlassian Jira, Confluence, Github
Research Methodologies	Physical Simulation, Bayesian Optimization, Statistical Validation, Clinical Trials

AWARDS & HONORS

2022 **Director's Medal**, Korean Information Society Development Institute

ACK

2022 **3rd Place**, Skin Burn Classification Hackathon

Seoul National University Hospital

2021 **Army Commendation Medal (ARCOM)**

U.S. Army